

Practical Demo: Hedging with Options on Treasury Futures

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This reading copy shows the calculations' results and figures. The HTML demo contains the full code in folded blocks and interactive scenario controls.

The desk question

What downside protection does a put overlay buy, and what remains exposed after paying its premium?

Learning Objectives

After completing this demo you should be able to read a Treasury futures option quote, separate intrinsic value from time value, calculate a Black futures-option benchmark, size a protective-put overlay with DV01, compare its scenario P&L with a linear futures hedge, and explain basis, volatility, and time-decay risks.

1 Inputs and Assumptions

The decision: insure a bond portfolio against rising yields without giving up all gains if yields fall. Follow **quote** → **model premium** → **implied volatility** → **hedge size** → **expiration payoff**. A put on Treasury futures protects against falling futures prices; that is the direction normally associated with rising yields.

All inputs are **illustrative, market-like inputs**, not live quotes. The option is treated as European for Black-model transparency; actual contract exercise and margin conventions must be checked. The bond portfolio is mapped to one Treasury futures contract with DV01; the CTD, conversion factor, basis, convexity, transaction costs, margin cash flows, and early exercise are held fixed or ignored.

Workflow stage	In this case
Observed or supplied	Futures, strike, quoted premium, expiry, and portfolio DV01
Estimated	Black benchmark value and futures-equivalent exposure
Diagnosed	Quote-versus-model gap, intrinsic value, and time value
Used for	Choosing between a linear hedge and asymmetric protection

2 Treasury Option Quote Convention

Treasury futures and their options are quoted in price points and fractions. For a \$100,000 face-value contract, one full point is \$1,000 and 1/64 of a point is \$15.625. A premium of 1-54 means

$$1 + \frac{54}{64} = 1.84375 \text{ points,}$$

or \$1,843.75 per contract. Price points, dollars, and yield basis points are different units.

```
Quoted premium (points)          1.8438
Quoted premium ($/contract)      1843.7500
dtype: float64
```

3 Intrinsic Value, Time Value, and Black's Model

For a futures put, intrinsic value is $\max(K - F, 0)$. Black's European futures-put formula is

$$p = e^{-rT} [KN(-d_2) - FN(-d_1)],$$

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2 T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

```
Black put value (points)          1.8531
Black put value ($/contract)      1853.0571
Intrinsic value (points)          0.0000
Time value (points)              1.8531
Illustrative quoted premium (points) 1.8438
dtype: float64
```

The quoted premium and model value need not match. A trader can interpret their difference as a signal to investigate volatility, exercise style, liquidity, model inputs, and bid-ask effects—not as an automatic arbitrage.

4 Size the Protective Put Overlay

4.1 Fit the quote, then use the fitted model

Instead of treating 7.5% volatility as the answer, invert the Black formula: find the volatility whose model premium equals the supplied 1.84375-point quote. This is an **implied** volatility under a European model, not a forecast of realized volatility. The quote remains illustrative.

```
Implied volatility (%)      7.4693
Repriced quote (points)    1.8438
Repricing error (points)   0.0000
dtype: float64
```

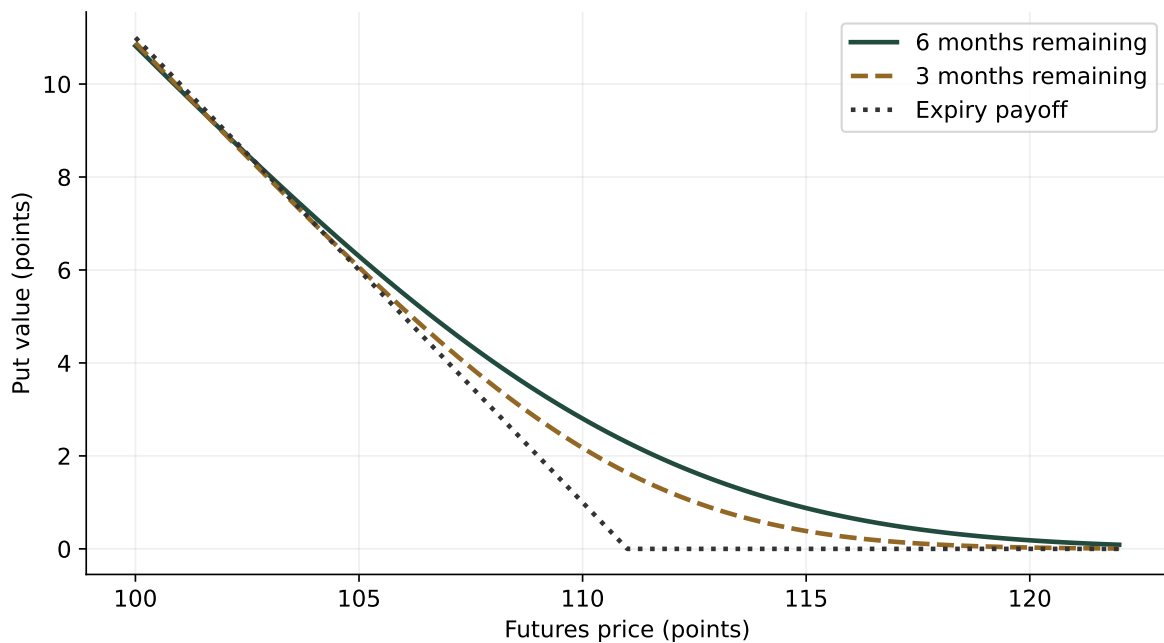


Figure 1: Before expiry the fitted model includes remaining optionality; at expiry only intrinsic payoff remains.

First map the portfolio into futures-equivalent contracts:

$$N_{fut-eq} = \frac{DV01_{portfolio}}{DV01_{futures}}.$$

For a one-for-one static floor at expiration, buy approximately one put per futures-equivalent contract. A current delta hedge would generally require a different count and rebalancing.

Exact futures equivalents	230.77
Static puts purchased	231.00
Premium cost (\$)	425906.25
Residual DV01 after rounding (\$/bp)	-18.00

dtype: float64

5 Futures-Price and Yield Scenarios

Approximate the futures-price change from a Treasury yield shock:

$$\Delta F \approx -\frac{DV01_{futures}}{\$1,000 \text{ per point}} \Delta y_{bp}.$$

The mapped bond P&L moves like a long futures position. The protective put pays only below the strike. A short-futures hedge offsets mapped price changes linearly in both directions.

	Yield shock (bp)	Terminal futures	Mapped bond P&L (\$)
0	-75	118	1,351,350
1	-40	115	720,720
2	0	112	0
3	40	109	-720,720
4	75	106	-1,351,350

	Put payoff (\$)	Bond + put P&L after premium (\$)	Bond + short futures P&L (\$)
0	0	925,444	0
1	0	294,814	0
2	0	-425,906	0
3	489,720	-656,906	0
4	1,120,350	-656,906	0

The put preserves gains when yields fall but requires an upfront premium. The linear futures hedge has no option premium in this simplified comparison, but it gives up gains as well as losses.

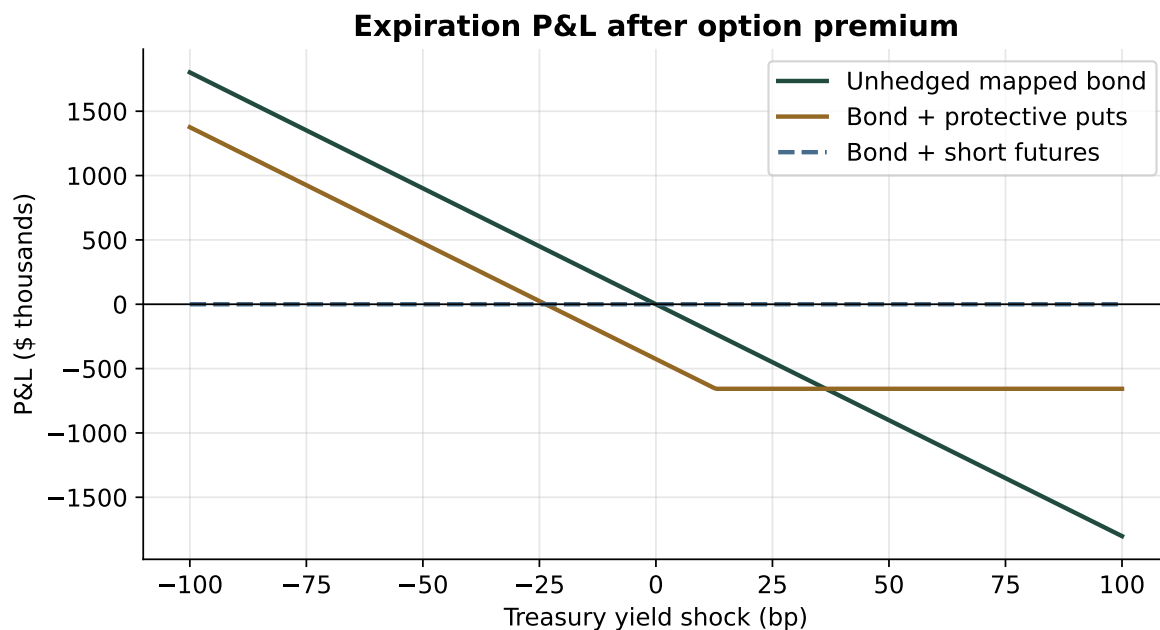


Figure 2: A protective put creates a floor; a short-futures hedge removes both upside and downside

6 Risks the Simple Comparison Omits

💡 Read the protection, not just the payoff

At an unchanged futures price, the out-of-the-money put expires worthless and the combined position loses the premium. When futures fall below 111, protection starts to offset further mapped bond losses. The floor is below zero because the manager bears both the move from 112 to 111 and the premium. A static expiry floor is not today's delta-neutral hedge, nor is it a guarantee for the actual bond portfolio if the futures mapping breaks down.

- **Basis risk:** the bond portfolio and Treasury futures may move differently.
- **Curve risk:** a single contract cannot hedge every maturity exposure.
- **Volatility risk:** option value changes when implied volatility changes.
- **Time decay:** the put loses time value as expiration approaches, all else equal.
- **Delta drift:** the option's sensitivity changes with price, time, and volatility.
- **Exercise and margin conventions:** listed contracts can differ from the European, premium-paid simplification.
- **Liquidity and roll risk:** bid-ask costs and hedge replacement affect realized results.

7 Validation Checks

Model put value exceeds intrinsic value	True
1-54 quote converts to \$1,843.75	True
Rounding residual is within half one contract DV01	True
Linear futures hedge offsets mapped P&L	True
Put improves the rising-yield outcome	True
dtype: bool	

8 Financial Interpretation

Contract selection begins with the risk, not the forecast. A short futures hedge is appropriate when the objective is to remove exposure linearly. A protective put is appropriate when the manager wants a loss floor while retaining favorable bond-price gains. That asymmetry is valuable, but its premium, volatility exposure, time decay, basis, and expiration must be budgeted explicitly.

8.1 Market handoff

Use [CME's Treasury options guide](#) to connect the case to listed contracts. Before replacing the supplied quote, record contract month, option expiry, strike, bid/ask, quotation unit, exercise style, and timestamp. Different contracts can use different tick conventions. Keep the embedded quote if comparable licensed data are unavailable. It is not a current exchange observation.

Your deliverable: compare premium paid, loss under a +75 bp yield shock, upside retained under -75 bp, and one reason the fitted European model may not match a listed option's valuation.

9 Interactive explorer

Change volatility while holding futures price, strike and interest rate fixed. Higher volatility makes the protection more expensive. Compare remaining maturities, then return to the fitted volatility reported above. The slider does not change the fixed premium already paid in the expiration-P&L example.

The HTML control selects a precomputed scenario. In the PDF, use the preceding figures and tables and rerun the displayed notebook inputs to explore other values.

10 Mini Project

Project brief. Use the demo's quoted premium and static contract count. Compare the actual portfolio (not just its rounded futures equivalent) with no hedge, with puts, and with short futures at expiry under -75, 0, and +75 bp shocks. Include the cost of funding the upfront premium at the risk-free rate. Report residual DV01 from contract rounding and explain why a terminal-payoff hedge is not a delta-neutral hedge today.

Deliverable: a one-page decision memo with a conventions statement, reproducible calculations, a scenario table, and a limitation that could change the decision.

Worked solution: calculations and decision memo

	Yield shock (bp)	Futures at expiry	Unhedged P&L (\$)
0	-75	117.85	1,350,000
1	0	112	-0
2	75	106.15	-1,350,000

	Put payoff (\$)	Put-protected P&L (\$)	Futures-hedged P&L (\$)
0	0	915,055.15	-1,350
1	0	-434,944.85	0
2	1,120,350	-664,594.85	1,350

```
Funded premium ($)      434944.85
Residual DV01 ($/bp)    -18.00
dtype: float64
```

The put pays only when the futures finish below strike. It preserves upside after premium cost, while short futures offset rate gains and losses symmetrically. The remaining cash-portfolio exposure equals the rounding residual under the constant-DV01 assumption. Before expiry, put delta has magnitude below one, so this static terminal protection is not a full current DV01 hedge. The European Black calculation is a teaching approximation for American-style Treasury options, and ignores exercise, basis, changing DV01, and trading frictions.